

(MIMS API Guide for 7 Modules)

MIMS INTEGRATED Modules

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2. [DRUG - DRUG ALERT](#)
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4. [DRUG - HEALTH ALERT](#)
5. [DRUG - DUPLICATE ALERT](#)
6. [DRUG - PREGNANCY ALERT](#)
 - a) [WOMEN OF CHILDBEARING AGE \(WOCBA\) ALERT](#)
 - b) [PREGNANCY ALERT](#)
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DOCUMENT REVISION CONTROL

Date	Version	Changes	Author(s)	Title
2023	1.0		Ashutosh Mishra	

INTENDED AUDIENCE

- Business and technical analysts.
- Application developers.

SCOPE OF THE DOCUMENT

This document provides an integration guide on MIMS Drug Information and CDS Alert to system integrator. It includes :

- the Request XML structure and input parameters
- Result XML structures and returned warnings

The system developer should have the experience of using FastTrack API to integrate the basic modules like Drug-Drug Interaction before he/she understands this document.

Sample Mapped Files

Hospital Formulary Master

Hospital's Item code	Hospital's Item name (Brand name)	CIMS DESC	CIMS GUID	CIMS Type
DurgCode1	SIMARC 2MG TAB	warfarin 2mg Oral Tablet	{C26C4DD1-7BB2-4CF7-9329-B3CFC1DBAC29}	GGPI
DurgCode2	ABBOTIC XL 500MG TAB	clarithromycin 500mg Oral Tablet	{48FAE690-3728-4A4B-BA04-90044160B80B}	GGPI
DurgCode3	LIPITOR 20MG TAB	Lipitor FC tab 20 mg	{0C8DA0BB-4E41-4274-927C-D0C2340BC7FE}	Product
DurgCode4	FLUCONAZOLE 150MG TAB	fluconazole 150mg Oral Tablet	{2F83FC84-9BB8-4D54-A87B-FBD925691FA1}	GGPI
DurgCode5	PRO TB 4 TAB	ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet	{EB883313-CB1A-4E33-93CE-C4C8A171124A}	GenericItem
DurgCode6	TEGRETOL 200MG TAB	Tegretol tab 200 mg	{D660DE2D-44B0-492F-98DE-9A6992D246AB}	Product
DurgCode7	in/ CLOPIDOGREL 75MG TAB	clopidogrel 75mg Oral Tablet	{850620BD-554E-4D31-9DDB-8BD6A444C1A7}	GGPI
DurgCode8	PLAVIX 75MG TAB	Plavix FC tab 75 mg	{67DD8810-FC34-4253-99EF-96B33AEA9B7B}	Product
DurgCode9	AMIODARONE HCL 50MG/ML-3ML	amiodarone hydrochloride 150mg per 3mL Injection	{B45F1C0B-BD66-2784-E034-080020E1DD8C}	GGPI
DurgCode10	CORDARONE 150MG/3ML INJ	Cordarone inj 150 mg/3 mL	{7211E99E-8A09-40CB-A20F-8FD840BEC083}	Product

Hospital Allergy Master

Hospital's Allergy Code	Hospital's Allergy Desc	CIMS DESC	CIMS GUID	CIMS Type
AllergyCode1	Atorvastatin	atorvastatin	{DAC45C7E-0C5E-42B5-9FB5-3527A1230370}	Molecule
AllergyCode2	HMG-CoA reductase inhibitors	HMG-CoA reductase inhibitors	{A1907026-49A8-4760-A9D1-BBC83F71212F}	SubstanceClass
AllergyCode3	Erythromycin	erythromycin	{4EB78A92-119E-4DE5-8F43-15548663ABB8}	Molecule
AllergyCode4	Amiodarone	Amiodarone	{722228FC-BBFF-473B-AD28-FF84B173CBDA}	SubstanceClass
AllergyCode5	warfarin	Warfarin	{4474F71F-26B0-411B-A3A2-16537132387F}	Molecule

DRUG INFORMATION

1. Monograph if a product is mapped to MIMS Product

```
<Request>
<Content>
<Product reference="{C8070B79-1627-4C0F-AC22-6DDA3DD0C5A9}"></Product>
</Content>
</Request>
```

2. Monograph if a product is mapped to MIMS GGPI

```
<Request>
<Content>
<GGPI reference="{0000423D-07F5-46E2-8A7A-D13C4C178A6A}">/>
</Content>
</Request>
```

3. Monograph if a product is mapped to MIMS GenericItem

```
<Request>
<Content>
<GenericItem reference="{025ABA8D-743D-4436-88CF-E15D2D51B616}">/>
</Content>
</Request>
```

The Display
of
Monograph
End User.
(Example)

Brief Full Generic	
Zithromax tab 250 mg	
Contents	Azithromycin dihydrate
Indications	<i>Tab/powd for oral susp</i> Lower resp tract infections including bronchitis & pneumonia; skin & soft tissue infections; acute otitis media; upper resp tract infections including sinusitis & pharyngitis/tonsillitis. Uncomplicated genital infections due to <i>Chlamydia trachomatis</i> & non-multiresistant <i>Neisseria gonorrhoea</i> . Chancroid due to <i>H ducreyi</i> . <i>Infusion</i> Community acquired pneumonia (CAP) & pelvic inflammatory disease (PID) in patients who require initial IV therapy.
Dosage and Administration	<i>Tab/powd for oral susp</i> Adult STD caused by <i>C trachomatis</i> or susceptible <i>N gonorrhoea</i> & chancroid due to <i>H ducreyi</i> 1 g as single dose. All other indications 500 mg daily for 3 days or 500 mg on day 1, then 250 mg on day 2-5. Childn Total dose of 30 mg/kg, given as single daily dose of 10 mg/kg daily for 3 days, or 10 mg/kg daily over 5 days w/ 10 mg/kg on day 1, then 5 mg/kg on days 2-5. Acute otitis media Alternative dosing is 30 mg/kg single dose. Ped Strep pharyngitis 10 or 20 mg/kg as single dose for 3 days. Max: 500 mg daily. Infusion Adult CAP 500 mg once daily for at least 2 days, followed by 500 mg PO once daily for 7 days. PID 500 mg once daily for 1-2 days, followed by 250 mg PO once daily for 7 days.
Contraindications	Hypersensitivity to azithromycin, erythromycin, other macrolides or ketolides.
Special Precautions	Discontinue if allergic reaction, signs & symptoms of hepatitis occur. Prolonged cardiac repolarization & QT interval. Diabetes or renal impairment. Superinfection or <i>Clostridium difficile</i> -associated diarrhea. Avoid co-administration w/ ergot derivatives. Pregnancy & lactation.
Adverse Drug Reactions	Mild neutropenia, hearing impairment, nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps), flatulence, abnormal liver function, allergic reactions including rash and angioedema. <i>Infusion</i> Local pain & inflammation at site of infusion.
Drug Interactions	Antacids, ergot derivatives. Monitor patients on concurrent warfarin, digoxin or cyclosporin.

DRUG-DRUG ALERT

Description	The Drug Alert module processes drug-drug interaction checks. An interaction warning displays essential information after checking for an interaction between two drugs.
Request XML Structure for Drug-Drug Alert	<pre> <Request> <Interaction> <Prescribing> <GGPI reference="{488F9F61-5D37-4989-925E-1742FFDAA9E}"/> <GGPI reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"/> <GGPI reference="{BF33752F-6062-0589-E034-0003BA299378}"/> <GGPI reference="{B3E696CF-71F8-6518-E034-080020E1DD8C}"/> </Prescribing> <References/> </Interaction> </Request> </pre> Current Medication Previous Medication

**Result XML
Structure
for Drug-
Drug Alert.**

```
<Result buildID="3310" copyright="Copyright (c) 2020 MIMS" expiryDate="2023-12-29">
  <Interaction>
    <GGPI name="captopril 100mg Oral Tablet" reference="{488F9F61-5D37-4989-925E-1742FFFDAA9E}">
    <Route name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}">
    <GGPI name="acetylsalicylic acid 100mg Enteric Coated Tablet" reference="{BF33752F-6062-0589-E034-0003BA299378}">
    <Route name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}">
    <ClassInteraction>
      <PrescribingInteractionClass description="ACE inhibitors inhibit the conversion of angiotensin I to angiotensin II."
        name="Angiotensin converting enzyme (ACE) inhibitors" reference="{97077317-8EB4-4EE1-8604-5654CF551B85}">
      <PrescribingMolecule name="captopril" reference="{A5A315C6-86E4-4ACE-89AF-E18B3AED1158}" />
    </PrescribingInteractionClass>
    <InteractionClass description="Analgesic, antipyretic and anti-inflammatory agents." name="Salicylates (systemic)"
      reference="{3CCDBD67-FD02-4AF8-B3EB-E54066A7BAD5}">
    <Molecule name="aspirin" reference="{68EC41A6-71EE-427C-B261-03425BAA3DD3}" />
    </InteractionClass>
    <Severity name="Moderate" ranking="4">Moderate</Severity>
    <Likelihood name="Good" ranking="3">Good</Likelihood>
    <Documentation name="Good" ranking="3">Good</Documentation>
    <Observation>
      <Abbreviated/>
      <Professional>has its effect reduced by</Professional>
      <Full/>
      <Consumer/>
    </Observation>
    <Interaction>
      <Abbreviated/>
      <Professional>Salicylates may reduce the beneficial effects of angiotensin converting enzyme inhibitors (ACEIs). Salicylates
        inhibit synthesis of prostaglandins and may antagonise antihypertensive effects of ACEIs in a dose-dependent manner, cause
        vasoconstriction/reduced cardiac output in heart failure patients, as well as increased risk of renal impairment. However,
        studies have shown that low-dose aspirin (less than or equals to 100 mg per day) is less likely to interact with ACEI.
        Furthermore, there are conflicting results of aspirin's effect to worsen heart failure in patients on concomitant ACEI.
        Nevertheless, if concurrent use is necessary, monitor patient's clinical condition, blood pressure and renal
        function.</Professional>
      <Full/>
      <Consumer/>
    </Interaction>
    <Precaution>
      <Abbreviated/>
      <Professional>Use combination with caution.</Professional>
      <Full/>
      <Consumer/>
    </Precaution>
    <Precaution>
      <Abbreviated/>
      <Professional>Monitor patient clinically.</Professional>
      <Full/>
      <Consumer/>
    </Precaution>
    <Precaution>
      <Abbreviated/>
      <Professional>Monitor renal function.</Professional>
      <Full/>
      <Consumer/>
    </Precaution>
    <Precaution>
      <Abbreviated/>
      <Professional>Monitor blood pressure.</Professional>
      <Full/>
      <Consumer/>
    </Precaution>
  </Interaction>
</Result>
```

```

</ClassInteraction>
</Route>
</GGPI>
<GGPI name="trimethoprim 300mg Oral Tablet" reference="{B3E696CF-71F8-6518-E034-080020E1DD8C}">
<Route name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}">
<ClassInteraction>
<PrescribingInteractionClass description="ACE inhibitors inhibit the conversion of angiotensin I to angiotensin II."
name="Angiotensin converting enzyme (ACE) inhibitors" reference="{97077317-8EB4-4EE1-8604-5654CF551B85}">
<PrescribingMolecule name="captopril" reference="{A5A315C6-86E4-4ACE-89AF-E18B3AED1158}"/>
</PrescribingInteractionClass>
<InteractionClass description="Folate antagonist antibacterial." name="Trimethoprim" reference="{14AA0897-6EB3-4003-
A3AF-262EDA2AE7A2}">
<Molecule name="trimethoprim" reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"/>
</InteractionClass>
<Severity name="Moderate" ranking="4">Moderate</Severity>
<Likelihood name="Good" ranking="3">Good</Likelihood>
<Documentation name="Good" ranking="3">Good</Documentation>
<Observation>
<Abbreviated>causes additive toxicity with</Abbreviated>
<Professional>causes additive toxicity with</Professional>
<Full>causes additional harmful effects with</Full>
<Consumer>causes additional harmful effects with</Consumer>
</Observation>
<Interaction>
<Abbreviated/>
<Professional>The concurrent use of ACE inhibitors and trimethoprim may lead to additive risk of hyperkalaemia by
inhibiting the renal excretion of potassium. Meta-analyses in geriatric population found that coadministration of co-
trimoxazole and an ACE inhibitor has been associated with higher risk of hospitalisation due to hyperkalaemia or sudden
death. Moreover, various reports of severe hyperkalaemia and arrhythmias have been described when trimethoprim or co-
trimoxazole was given concomitantly with an ACE inhibitor. Other predisposing factors include renal impairment, elderly,
high dosages and diabetes. Closely monitor at-risk patient's serum potassium levels and ECG during combined use. Consider
adjusting dose or use an alternative antibiotic in elderly patients.</Professional>
<Full/>
<Consumer/>
</Interaction>
<Precaution>
<Abbreviated/>
<Professional>Use combination with extreme caution.</Professional>
<Full/>
<Consumer/>
</Precaution>
<Precaution>
<Abbreviated/>
<Professional>Monitor serum potassium.</Professional>
<Full/>
<Consumer/>
</Precaution>
</ClassInteraction>
</Route>
</GGPI>
</Route>
</GGPI>
<GGPI name="amoxicillin trihydrate 250mg Oral Capsule" reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"/>
<GGPI name="acetylsalicylic acid 100mg Enteric Coated Tablet" reference="{BF33752F-6062-0589-E034-0003BA299378}"/>
<GGPI name="trimethoprim 300mg Oral Tablet" reference="{B3E696CF-71F8-6518-E034-080020E1DD8C}"/>
<DuplicateTherapy/>
</Interaction>
</Result>

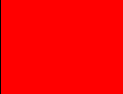
```

The
Display of
Drug - Drug
Alert to
End User.
(Example)

Drug Interaction (2)

- 
 captopril 100mg Oral Tablet (captopril/Systemic) vs trimethoprim 300mg Oral Tablet (trimethoprim/Systemic) [Use combination with extreme caution.]
- 
 captopril 100mg Oral Tablet (captopril/Systemic) vs acetylsalicylic acid 100mg Enteric Coated Tablet (aspirin/Systemic) [Use combination with caution.]

Drug Interaction Alert Severity Levels

Severity Level	Colour	Definition
Severe		The interaction between these medications may be life-threatening or may cause permanent damage. These medications are not usually used concurrently; medical intervention may be required.
Moderate		These medications may interact resulting in the potential deterioration of the patient's condition. The patient should be monitored for the possible manifestations of the interaction. Medical intervention or a change in therapy may be required.
Minor		The interaction may occur based on the mechanism of action of the co-administered medicines. Be alert for increased or decreased effect, depending on the combination of medicines.
Caution		Clinical effects of the interaction are limited and may be bothersome but would not usually require a major change to therapy. The patient should be monitored for the possible manifestations of the interaction.

Documentation Levels

Documentation Level	Definition
Well established	There have been several published reports of this interaction. The pharmacological explanation of why the interaction occurs is well documented and understood. There are usually controlled studies that have established that the interaction exists.
Good	Although controlled studies may not have been performed, several case reports have been documented and other data strongly suggests this interaction exists.
Limited	Few reports of this interaction exist. These few reports usually consist of limited case reports where clinically sound justification of the interaction is found.
Not established	The interaction may have occurred with other medicines within the same class, or there is a theoretical possibility that the interaction exists.

DRUG-ALLERGY ALERT

Drug Allergy Alert	Prescribing or Prescribed Drug - Product - Generic product - Generic Item Existing Drug Allergies - Product - Generic product - Generic Item - Active Composition Group - Molecule - Substance Class	- Adalat 5mg capsule / nifedipine 5mg capsule / nifedipine capsule - Abilify tab 15 mg - fluorouracil 500mg Injection - nifedipine capsule - benzoyl peroxide – adapalene - fluorouracil - Cyanoacrylates
--------------------	---	---

XML Request and Response for Allergy Alert

Request XML Structure for Drug-Allergy Alert.

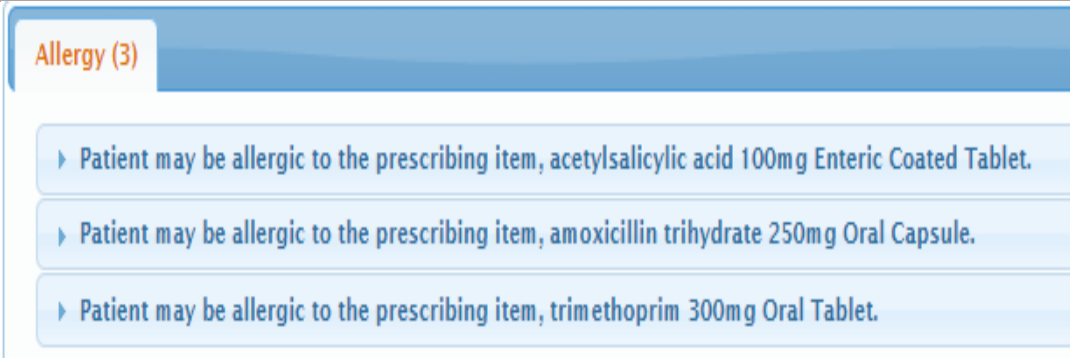
```

<Request>
<Interaction>
<Prescribing>
<GGPI reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"/>
<GGPI reference="{BF33752F-6062-0589-E034-0003BA299378}"/>
<GGPI reference="{B3E696CF-71F8-6518-E034-080020E1DD8C}"/>
</Prescribing>
<Allergies>
<Molecule reference="{5143B2B1-DBED-45A2-8308-61636C3A2F0B}"/>
<Molecule reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"/>
<Molecule reference="{74DCDDAF-9349-430F-A2F2-B7F2BC5CD8AA}"/>
<SubstanceClass reference="{8B9ECB52-CDC9-4460-AB40-6E3C7DF38B1E}"/>
<SubstanceClass reference="{9E775009-231D-728D-E034-080020E1DD8C}"/>
</Allergies>
<References/>
</Interaction>
</Request>

```

You need to add the Drugs in

You have to register Allergy inside the Allergies tag

Result XML Structure for Drug-Allergy Alert	<pre> <Result buildID="3310" copyright="Copyright (c) 2020 MIMS" expiryDate="2023-12-29"> <Interaction> <GGPI name="amoxicillin trihydrate 250mg Oral Capsule" reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"> <Allergy> <SubstanceClass name="Penicillins" reference="{9E775009-231D-728D-E034-080020E1DD8C}"> <PrescribingMolecule name="amoxicillin trihydrate" reference="{5504533E-4E7C-4862-98B3-076801506CA1}"/> </SubstanceClass> </Allergy> </GGPI> <GGPI name="acetylsalicylic acid 100mg Enteric Coated Tablet" reference="{BF33752F-6062-0589-E034-0003BA299378}"> <Allergy> <SubstanceClass name="Salicylates" reference="{8B9ECB52-CDC9-4460-AB40-6E3C7DF38B1E}"> <PrescribingMolecule name="aspirin" reference="{68EC41A6-71EE-427C-B261-03425BAA3DD3}"/> </SubstanceClass> </Allergy> </GGPI> <GGPI name="trimethoprim 300mg Oral Tablet" reference="{B3E696CF-71F8-6518-E034-080020E1DD8C}"> <Allergy> <Molecule name="trimethoprim" reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"> <SubstanceClass name="Trimethoprim and related agents" reference="{59E6DC7E-CD79-4629-8E98-42E1D56B6B06}"> <PrescribingMolecule name="trimethoprim" reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"/> </SubstanceClass> </Molecule> </Allergy> </GGPI> <DuplicateTherapy/> </Interaction> </Result> </pre>
The Display of Drug-Allergy Alert to End User. (Example)	

Drug Allergy Interaction Types

Allergy Alert Type	Short Description	Full Definition
Basic Allergy Alert:	Same ingredient as allergen	Patient may be allergic to the prescribed medicine containing the same ingredient as patient's documented allergen.
Substance Class Allergy Alert:	Same allergy class	Patient may be allergic to the prescribed medicine that contains one or more molecules belonging to the same allergy class as patient's documented allergen.
Cross-Sensitivity Allergy Alert:	Cross-sensitivity	Patient may be allergic to the prescribed medicine containing one or more molecules that has cross sensitivity with patient's documented allergen.

DRUG-HEALTH ALERT

MIMS Alerts	Parameters needed	Examples
Drug Health Alert	- Prescribing Drug (Generic Product / Generic Item) - Patient Health Condition in ICD10 code	- nifedipine 5mg capsule / nifedipine capsule - H36.0 (Diabetic retinopathy) / N17 (Acute renal failure)

XML Request and Response for Drug to Health Alert

Description	The Drug Health Alert module is used in conjunction with the patient's profile for stored medical conditions and subsequently for potential contraindications with the prescribed medication. The database currently supports ICD10.
Request XML Structure for Drug Health Alert check.	<pre> <Request> <Interaction> <Prescribing> <GGPI reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"/> <GGPI reference="{BF33752F-6062-0589-E034-0003BA299378}"/> <GGPI reference="{B3E696CF-71F8-6518-E034-080020E1DD8C}"/> </Prescribing> <HealthIssueCodes> <HealthIssueCode code="J45" codeType="ICD10"/> <HealthIssueCode code="N17" codeType="ICD10"/> </HealthIssueCodes> <References/> </Interaction> </Request> </pre> Current and Past Medications Patient's Disease Conditions

**Result
XML
Structure
for Drug
Health
Alert
check.**

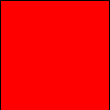
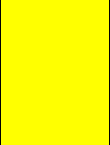
```
<Result buildID="3310" copyright="Copyright (c) 2020 MIMS" expiryDate="2023-12-29">
  <Interaction>
    <GGPI name="amoxicillin trihydrate 250mg Oral Capsule" reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"/>
    <GGPI name="acetylsalicylic acid 100mg Enteric Coated Tablet" reference="{BF33752F-6062-0589-E034-0003BA299378}"/>
    <Route name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"/>
    <HealthIssueCode code="J45" codeType="ICD10" name="Asthma">
      <ClassInteraction>
        <PrescribingInteractionClass description="Analgesic, antipyretic and anti-inflammatory agents." name="Salicylates"
          reference="{9EE10324-3553-44BE-A6D4-14419B7ADEA7}"/>
        <PrescribingMolecule name="aspirin" reference="{68EC41A6-71EE-427C-B261-03425BAA3DD3}"/>
      </PrescribingInteractionClass>
      <Severity name="Extreme Caution" ranking="2"/>
      <Likelihood name="Limited" ranking="2">Limited</Likelihood>
      <Documentation name="Limited" ranking="2">Limited</Documentation>
      <Interaction>
        <Abbreviated>Salicylates may precipitate or exacerbate asthma, and risk increases in asthmatic patients with chronic rhinitis or a history of nasal polyps. Contraindicated in severe asthma.</Abbreviated>
        <Professional>Salicylates may precipitate or exacerbate asthma, and risk increases in asthmatic patients with chronic rhinitis or a history of nasal polyps. Contraindicated in severe asthma.</Professional>
      <Full/>
    </Consumer/>
  </Interaction>
</ClassInteraction>
</HealthIssueCode>
<HealthIssueCode code="N17" codeType="ICD10" name="Acute renal failure">
  <ClassInteraction>
    <PrescribingInteractionClass description="Analgesic, antipyretic and anti-inflammatory agents." name="Salicylates"
      reference="{9EE10324-3553-44BE-A6D4-14419B7ADEA7}"/>
    <PrescribingMolecule name="aspirin" reference="{68EC41A6-71EE-427C-B261-03425BAA3DD3}"/>
  </PrescribingInteractionClass>
  <Severity name="Contraindicated" ranking="3"/>
  <Likelihood name="Good" ranking="3">Good</Likelihood>
  <Documentation name="Good" ranking="3">Good</Documentation>
  <Interaction>
    <Abbreviated>Salicylates are associated with renal toxicity (e.g. nephropathy) and aspirin has been shown to prolong bleeding time in some patients with uraemia. Contraindicated in severe renal impairment.</Abbreviated>
    <Professional>Salicylates are associated with renal toxicity (e.g. nephropathy) and aspirin has been shown to prolong bleeding time in some patients with uraemia. Contraindicated in severe renal impairment.</Professional>
  <Full/>
  </Consumer/>
</Interaction>
</ClassInteraction>
</HealthIssueCode>
</Route>
</GGPI>
<DuplicateTherapy/>
</Interaction>
</Result>
```

The
Display of
Drug -
Health
Alert to
End User.
(Example)

Health Conditions (2)

- ▶  acetylsalicylic acid 100mg Enteric Coated Tablet (aspirin) vs N17 (Acute renal failure)
- ▶  acetylsalicylic acid 100mg Enteric Coated Tablet (aspirin) vs J45 (Asthma)

Drug Health Interaction Alert Severity Levels

Severity Level	Colour	Definition
Contraindicated		Absolute contraindication; the use of the drug is completely inadvisable and should be avoided.
Extreme Caution		Use of the drug is not recommended and the patient has a higher risk of developing adverse effects or exacerbation of their existing health condition. However, use is still possible and the risk may be outweighed by other considerations.

Documentation Levels

Documentation Level	Definition
Well established	There have been several published reports of this interaction. The pharmacological explanation of why the interaction occurs is well documented and understood. There are usually controlled studies that have established that the interaction exists.
Good	Although controlled studies may not have been performed, several case reports have been documented and other data strongly suggests this interaction exists.
Limited	Few reports of this interaction exist. These few reports usually consist of limited case reports where clinically sound justification of the interaction is found.
Not established	The interaction may have occurred with other medicines within the same class, or there is a theoretical possibility that the interaction exists.

DRUG-DUPLICATE ALERT

MIMS Alerts	Parameters needed	Examples
Duplicate Alert	Prescribing Drug - Generic Product - Generic Item	- nifedipine 5mg capsule - nifedipine capsule

XML Request and Response for Duplicate Alert

Description	Duplicate Alert Module provides an alert if the medicine to be prescribed, dispensed or administered has an identical therapeutic function to another medicine already prescribed to the patient.
Request XML Structure for Duplicate Alert check.	<pre> <Request> <Interaction> <Prescribing> <GGPI reference="{AA0D6353-F6E1-4AB1-AF1E-5E5F109A0706}"/> <GGPI reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}"/> <GGPI reference="{B7AD6ED9-C22D-561C-E034-080020E1DD8C}"/> </Prescribing> <References> <DuplicateTherapy checkSameDrug="true" /> <DuplicateIngredient checkSameDrug="true" /> </Interaction> </Request> </pre> Current and Past Medications You must add DuplicateIngredient and DuplicateTherapy tags

Result XML Structure for Duplicate Alert check.	<pre> <Result buildID="2992" copyright="Copyright (c) 2017 MIMS" expiryDate="2019-06-30"> <Interaction> <GGPI reference="{AA0D6353-F6E1-4AB1-AF1E-5E5F109A0706}" name="acidovir 800mg Film Coated Tablet" > </GGPI> <GGPI reference="{B3E6B75E-9519-6AE7-E034-080020E1DD8C}" name="amoxicillin trihydrate 250mg Oral Capsule" > </GGPI> <GGPI reference="{B7AD6ED9-C22D-561C-E034-080020E1DD8C}" name="acidovir 800mg Oral Tablet" > </GGPI> <DuplicateTherapy> <Warning Level="1"> <Duplicate ATCCode="J05AB01"> <GGPI reference="{AA0D6353-F6E1-4AB1-AF1E-5E5F109A0706}"/> <GGPI reference="{B7AD6ED9-C22D-561C-E034-080020E1DD8C}"/> <RouteOfAdministration name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"/> </Duplicate> </Warning> </DuplicateTherapy> <DuplicateIngredient> <Warning Level="1"> <Molecule reference="{44DCA166-6654-4E84-AEC1-C514DF40D9DE}" name="acidovir" > <GGPI reference="{AA0D6353-F6E1-4AB1-AF1E-5E5F109A0706}"/> <GGPI reference="{B7AD6ED9-C22D-561C-E034-080020E1DD8C}"/> </Molecule> <RouteOfAdministration name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"/> </Warning> </DuplicateIngredient> </Interaction> </Result> </pre> Returns that these two drugs having a Duplicate Therapy Alert Returns that these two drugs having a Duplicate Ingredient Alert
--	---

The Display of Duplicate Alert to End User.
(Example)

Duplicate Therapy (1)

Duplicate Ingredient (1)

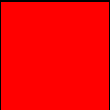
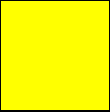
 aciclovir 800mg Film Coated Tablet and aciclovir 800mg Oral Tablet [ATC Code: J05AB01]

Duplicate Therapy (1)

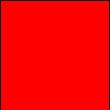
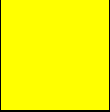
Duplicate Ingredient (1)

 aciclovir 800mg Film Coated Tablet and aciclovir 800mg Oral Tablet [aciclov

Duplicate Ingredient (DI) Alert Severity Levels

Severity Level	Colour	Definition
1		Same Ingredient in drug being Prescribed (exact match - i.e. identical)
2		This means prescribed drugs belong to same molecule family - i.e. other salts of the molecule (e.g. amlodipine besilate and amlodipine)
3		Algorithm checks across the molecule ancestry of grandparent, parent and child e.g. insulin (grandfather) >> insulin aspart (parent) >> insulin aspart protamine (child)

Duplicate Therapy (DT) Alert Severity Levels

Severity Level	Colour	Definition
1		full match of entire ATC code up to 5th level or 7 alphanumeric codes; drugs of identical formulation should have the same ATC code.(ATC code: N02BE01)
2		Partial match of ATC code up to 4th level or first 5 alphanumeric codes; drugs that belong to the same substance class. (ATC code: A10BB)
3		Partial match of ATC code up to 3rd level or first 4 alphanumeric codes; drugs with similar therapeutic use. (ATC code: A10B)

DRUG TO PREGNANCY ALERT

1. PREGNANCY ALERT FOR WOMAN OF CHILDBEARING AGE (WOCBA)

WOCBA ALERT

MIMS Alerts	Parameters needed	Examples
Woman Childbearing Age Alert	Prescribing Drug - Product - Generic Product - Generic Item Patient Information - Gender (Female) - Age (Day/Month/Year)	- Adalat 5mg capsule - nifedipine 5mg capsule - nifedipine capsule - Female (F) 25 (Years)

Examples for Pregnancy Parameters for Women Of Child Bearing Age(WOCBA) which you can choose:

Patient Age:

<Age>
 <Year>34</Year>
 <Month>8</Month>
 <Day>11</Day>
 </Age>

Gender:

Female:
 <Gender>F</Gender>

Male:
 <Gender>M</Gender>

XML Request and Response for WOCBA Alert

Description	If the input patient profile is a <u>Female and age between 15 to 44 years old</u> , WOCBA check may return a warning.
Request XML Structure of WOCBA Check.	<pre> <Request> <Interaction> <Prescribing> <GGPI reference="{C1DDD75F-FD9D-535C-E034-0003BA299378}"> </GGPI> </Prescribing> </Interaction> <PatientProfile> <Gender>F</Gender> <Age> <Day>26</Day> <Month>8</Month> <Year>35</Year> </Age> </PatientProfile> </Request> </pre> Patient is a Female and 35 years old in the Request XML

Result XML Structure of WOCBA Check.

```
<Result buildID="2992" copyright="Copyright (c) 2017 MIMS" expiryDate="2019-06-30">
<Interaction>
<GGPI reference="{C1DDD75F-FD9D-535C-E034-0003BA299378}" name="idarubicin hydrochloride 10mg Oral Capsule" >
<Route name="Oral" reference="{AD02A5E3-6437-12B1-E034-080020E1DD8C}">
<WOCBA>
<InteractionClass name="Idarubicin" reference="{97029F01-B65E-4F06-B8CC-4055932CBC66}">
<Molecule name="idarubicin hydrochloride" reference="{F8A20581-54AC-499F-BAE3-E80356E8DF87}" />
</InteractionClass>
<Category name = "D" Source = "FDA">
<Comment/>
</Category>
<Category name = "+" Source = "MIMS">
<Comment>Drug associated with adverse events in fetus. Instruct patient not to get pregnant or to use contraception. Administer only where highly unlikely to conceive and patient has been informed of potential hazards.</Comment>
</Category>
</WOCBA>
</Route>

</GGPI>
<DuplicateTherapy/>
<DuplicateIngredient/>
</Interaction>
</Result>
```

WOCBA check may return with a <WOCBA> tag in the Result XML with FDA, MIMS and ADEC comments.

The Display of WOCBA Alert to End User. (Example)

Pregnancy (1)

▼ idarubicin hydrochloride 10mg Oral Capsule (idarubicin hydrochloride/Oral) [Women of Childbearing Age]

FDA Category

A B C **D** X

There is positive evidence of human fetal risk based on adverse reaction data from investigational or marketing experience or studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks.

MIMS Notes

Drug associated with adverse events in fetus. Instruct patient not to get pregnant or to use contraception. Administer only where highly unlikely to conceive and patient has been informed of potential hazards.

WOMAN OF CHILDBEARING AGE Severity Levels

Severity Level	Colour	Definition
X	Red	Studies in animals or human beings have demonstrated foetal abnormalities or there is evidence of foetal risk based on human experience or both, and the risk of the use of the drug in pregnant women clearly outweighs any possible benefit. The drug is contraindicated in women who are or may become pregnant.
D	Orange	There is positive evidence of human foetal risk, but the benefits from use in pregnant women may be acceptable despite the risk (e.g., if the drug is needed in a life-threatening situation or for a serious disease for which safer drugs cannot be used or are ineffective).
C	Yellow	Either studies in animals have revealed adverse effects on the foetus (teratogenic or embryocidal or other) and there are no controlled studies in women or studies in women and animals are not available. Drugs should be given only if the potential benefit justifies the potential risk to the foetus.
B	Light Green	Either animal-reproduction studies have not demonstrated a foetal risk but there are no controlled studies in pregnant women or animal-reproduction studies have shown an adverse effect (other than a decrease in fertility) that was not confirmed in controlled studies in women in the 1st trimester (and there is no evidence of a risk in later trimesters).
A	Blue	Controlled studies in women fail to demonstrate a risk to the foetus in the 1st trimester (and there is no evidence of a risk in later trimesters), and the possibility of foetal harm remains remote.
+	Yellow	MIMS Notes

2. PREGNANCY ALERT FOR PREGNANT WOMAN CONSIDERING THE ACTUAL PREGNANCY DURATION

PREGNANCY ALERT

MIMS Alerts	Parameters needed	Examples
Pregnancy Alert	Prescribing Drug - Product - Generic Product - Generic Item Patient Information - Gender (Female) - Age (Day/Month/Year) - Pregnancy period (Day/Week/Month) - Pregnancy (true)	- Adalat 5mg capsule - nifedipine 5mg capsule - nifedipine capsule - Female (F) 25 (Years) 20 (Weeks)

Examples for Pregnancy Parameters which you can choose:		
Pregnancy Duration: By Month ----- <Pregnancy> <Month>4</Month> </Pregnancy> OR By Week <Pregnancy> <Week>4</Week> </Pregnancy> OR By Day <Pregnancy> <Day>140</Day> </Pregnancy> OR Considering all Parameters together ----- <Pregnancy> <Month>2</Month> <Week>3</Week> <Day>5</Day> </Pregnancy>	Patient Age: <Age> <Year>34</Year> <Month>8</Month> <Day>11</Day> </Age>	Gender: Female: <Gender>F</Gender> Male: <Gender>M</Gender>

XML Request and Response for Pregnancy Alert

Description	If the input patient profile is a <u>Female with pregnancy period</u>, Pregnancy check will return a warning if there is any. For ADEC and FDA Pregnancy Category description, please refer to website http://en.wikipedia.org/wiki/Pregnancy_category.
--------------------	---

Request XML Structure for Pregnancy check.	<pre> <Request> <Interaction> <Prescribing> <GGPI reference="{C1DDD75F-FD9D-535C-E034-0003BA299378}"> <RouteOfAdministration name="Intravenous"/> </GGPI> </Prescribing> <References/> </Interaction> <PatientProfile> <Gender>F</Gender> <Age><Year>34</Year></Age> <Pregnancy><Week>20</Week></Pregnancy> </PatientProfile> </Request> </pre> You need to add the Patient Profile tag Patient is in 20 weeks (2nd Trimester) pregnancy
Result XML Structure for Pregnancy check.	<pre> <Result buildID="2992" copyright="Copyright (c) 2017 MIMS" expiryDate="2019-06-30"> <Interaction> <GGPI reference="{C1DDD75F-FD9D-535C-E034-0003BA299378}" name="idarubicin hydrochloride 10mg Oral Capsule" > <Route name="Oral" reference="{AD02A5E3-6437-12B1-E034-080020E1DD8C}"> <Pregnancy> <InteractionClass name="Idarubicin" reference="{97029F01-B65E-4F06- B8CC-4055932CEB66}"> <Molecule name="idarubicin hydrochloride" reference="{F8A20581-54AC- 499F-BAE3-E80356E8DF87}"> </InteractionClass> <Category name = "D" Trimester = "2nd" Source ="FDA"> <Comment/> </Category> <Category name = "+" Trimester = "2nd" Source ="MIMS"> <Comment>Experts cite that scientific literature suggests a possible association of occupational exposure to certain cytotoxic agents during the first trimester of pregnancy with fetal loss or malformation.</Comment> <Category> </Pregnancy> </Route> </GGPI> <DuplicateTherapy/> <DuplicateIngredient/> </Interaction> </Result> </pre> Pregnancy check returns warnings under <Pregnancy> tag with FDA, MIMS and ADEC comments.

The Display of Pregnancy Alert to End User. (Example)

Pregnancy (1)

▼ idarubicin hydrochloride 10mg Oral Capsule (idarubicin hydrochloride/Oral) [2nd Trimester]

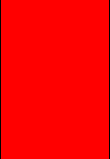
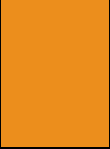
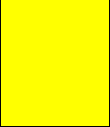
FDA Category
A
B
C
D
X

There is positive evidence of human fetal risk based on adverse reaction data from investigational or marketing experience or studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks.

MIMS Notes

Experts cite that scientific literature suggests a possible association of occupational exposure to certain cytotoxic agents during the first trimester of pregnancy with fetal loss or malformation.

ACTUAL PREGNANCY Alert Severity Levels

Severity Level	Colour	Definition
X		Studies in animals or human beings have demonstrated foetal abnormalities or there is evidence of foetal risk based on human experience or both, and the risk of the use of the drug in pregnant women clearly outweighs any possible benefit. The drug is contraindicated in women who are or may become pregnant.
D		There is positive evidence of human foetal risk, but the benefits from use in pregnant women may be acceptable despite the risk (e.g., if the drug is needed in a life-threatening situation or for a serious disease for which safer drugs cannot be used or are ineffective).
C		Either studies in animals have revealed adverse effects on the foetus (teratogenic or embryocidal or other) and there are no controlled studies in women or studies in women and animals are not available. Drugs should be given only if the potential benefit justifies the potential risk to the foetus.
B		Either animal-reproduction studies have not demonstrated a foetal risk but there are no controlled studies in pregnant women or animal-reproduction studies have shown an adverse effect (other than a decrease in fertility) that was not confirmed in controlled studies in women in the 1st trimester (and there is no evidence of a risk in later trimesters).
A		Controlled studies in women fail to demonstrate a risk to the foetus in the 1st trimester (and there is no evidence of a risk in later trimesters), and the possibility of foetal harm remains remote.
+		MIMS Notes

DRUG TO LACTATION ALERT

MIMS Alerts	Parameters needed	Examples
Lactation Alert	Prescribing Drug - Product - Generic Product - Generic Item Patient Information - Gender (Female) - Nursing (true)	- Adalat 5mg capsule - nifedipine 5mg capsule - nifedipine capsule - Female (F) - true

Examples for Pregnancy Parameters for Women Of Child Bearing Age (WOCBA) which you can choose:

Patient Age:	Gender:	Nursing:
<Age> <Year>34</Year> <Month>8</Month> <Day>11</Day> </Age>	Female: <Gender>F</Gender> Male: <Gender>M</Gender>	<Nursing>true</Nursing> <Nursing>false</Nursing>

XML Request and Response for Lactation Alert

Description	If the input patient profile is a <u>Female with "True" Nursing flag</u> , Lactation check will return a warning if there is any.
Request XML Structure for Lactation check.	<pre> <Request> <Interaction> <Prescribing> <GGPI reference="{2AA7FC83-2E12-496E-98A8-5F0B8D548907}"> <RouteOfAdministration name="Intramuscular"/> </GGPI> </Prescribing> <References/> </Interaction> <PatientProfile> <Gender>F</Gender> <Age><Year>34</Year></Age> <Nursing>true</Nursing> </PatientProfile> </Request> </pre> Patient is a nursing mother (Lactation). You need to add the patient profile tag as well as Nursing tag inside it.
Result XML Structure for Lactation check.	<pre> <Result buildID="2992" copyright="Copyright (c) 2017 MIMS" expiryDate="2019-06-30"> <Interaction> <GGPI reference="{2AA7FC83-2E12-496E-98A8-5F0B8D548907}" name="cimetidine 200mg Caplet" > <Route name="Systemic" reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"> <Lactation> <InteractionClass name="Cimetidine" reference="{F78FE8DE-F39D-4578-BF39-C85AB23307AE}"> <Molecule name="cimetidine" reference="{F5BCEAB2-209E-411A-BD71-FF75EAA87D9B}"/> </InteractionClass> <Severity name="Caution" ranking="3"> </Severity> <Comment>Cimetidine is excreted in human milk but there are no reported adverse effects on lactating infants. However, there is a theoretical risk for inhibition of gastric acidity, inhibition of drug metabolism and stimulation of the central nervous system in the nursing infant. Moreover, data on long-term side effects is lacking.</Comment> <References> <ElectronicReference> <Title>Cimetidine</Title> <RefTitle>Drugs in Pregnancy & Lactation: A Reference Guide to Fetal and Neonatal Risk, 8th Edition</RefTitle> <SystemName> </SystemName> <SystemAuthor>Briggs GG, Freeman RK, Yaffe SJ</SystemAuthor> <SystemManf>Lippincott Williams & Wilkins, A Wolters Kluwer business</SystemManf> <SystemManfLoc>Philadelphia, PA</SystemManfLoc> <SystemManfCountry>USA</SystemManfCountry> <URL>www.briggsdrugsinpregnancy.com/</URL> <Date>04/02/2010</Date> </ElectronicReference> </pre>

	<pre> <WebReference> <Author>World Health Organization; UNICEF</Author> <Title>Breastfeeding and maternal medication - Recommendations for drugs in the eleventh WHO model list of essential drugs</Title> <Site> </Site> <URL>http://www.who.int/</URL> <Date>2002</Date> </WebReference> </References> </Lactation> </Route> </GGPI> </Interaction> </Result> </pre>
The Display of Lactation Alert to End User. (Example)	Lactation (1) ▼ cimetidine 200mg Caplet Severity Level <i>Caution</i> - These drugs may theoretically cause side effects to a nursing infant but have either not been observed to do so or have only occasionally caused mild side effects. These agents may be given to a lactating mother but monitoring is needed for possible toxicity to either mother or infant. Description Cimetidine is excreted in human milk but there are no reported adverse effects on lactating infants. However, there is a theoretical risk for inhibition of gastric acidity, inhibition of drug metabolism and stimulation of the central nervous system in the nursing infant. Moreover, data on long-term side effects is lacking. References - Cimetidine. Drugs in Pregnancy & Lactation: A Reference Guide to Fetal and Neonatal Risk, 8th Edition. . Briggs GG, Freeman RK, Yaffe SJ. Lippincott Williams & Wilkins, A Wolters Kluwer business. Philadelphia, PA. USA. Available from URL: www.briggsdrugsinpregnancy.com/. (Accessed on 04/02/2010) - Breastfeeding and maternal medication - Recommendations for drugs in the eleventh WHO model list of essential drugs. . URL: http://www.who.int/. (Accessed on 2002)

Drug Lactation Alert Severity Levels

Severity Level	Colour	Definition
Contraindicated	Red	Human experience may or may not be available but these drugs can have severe toxicity to a nursing infant. The drug should not be given to a nursing mother. These drugs also include those that are indicated in conditions where lactation is a contraindication.
Avoid if possible	Orange	These drugs have been reported to cause clinically significant side effects to a nursing infant, and/or existing animal data suggest significant risks to a nursing infant. These agents may also inhibit lactation or may harm the mother. If use during breastfeeding is unavoidable, monitoring is needed for possible toxicity to either mother or infant.
Caution	Yellow	These drugs may theoretically cause side effects to a nursing infant but have either not been observed to do so or have only occasionally caused mild side effects. These agents may be given to a lactating mother but monitoring is needed for possible toxicity to either mother or infant.

Documentation Levels

Documentation Level	Definition
Well established	There have been several published reports of this interaction. The pharmacological explanation of why the interaction occurs is well documented and understood. There are usually controlled studies that have established that the interaction exists.
Good	Although controlled studies may not have been performed, several case reports have been documented and other data strongly suggests this interaction exists.
Limited	Few reports of this interaction exist. These few reports usually consist of limited case reports where clinically sound justification of the interaction is found.
Not established	The interaction may have occurred with other medicines within the same class, or there is a theoretical possibility that the interaction exists.

Full Query

Now you pick the corresponding “MIMS TYPE” and “MIMS GUID” and build the “request.xml” as Below

“Request.xml”

```
<Request>
<Interaction>
<Prescribing>
<GGPI reference="{850620BD-554E-4D31-9DD8-8BD6A444C1A7}" />
<Product reference="{67DD8810-FC34-4253-99EF-96B33AEA9B7B}" />
<GGPI reference="{B45F1C0B-BD66-2784-E034-080020E1DD8C}" />
<Product reference="{7211E99E-8A09-40CB-A20F-8FD840BEC083}" />
<GGPI reference="{C26C4DD1-7BB2-4CF7-9329-B3CFC1DBAC29}" />
<Product reference="{48FAE690-3728-4A4B-BA04-90044160B80B}" />
<GGPI reference="{0C8DA0BB-4E41-4274-927C-D0C2340BC7FE}" />
<GGPI reference="{2F83FC84-9BB8-4D54-A87B-FBD925691FA1}" />
<GenericItem reference="{EB883313-CB1A-4E33-93CE-C4C8A171124A}" />
<Product reference="{D660DE2D-44B0-492F-98DE-9A6992D246AB}" />
</Prescribing>
<Allergies>
<Molecule reference="{DAC45C7E-0C5E-42B5-9FB5-3527A1230370}" />
<SubstanceClass reference="{A1907026-49A8-4760-A9D1-BBC83F71212F}" />
<Molecule reference="{4EB78A92-119E-4DE5-8F43-15548663ABB8}" />
<SubstanceClass reference="{722228FC-BBFF-473B-AD28-FF84B173CBDA}" />
<Molecule reference="{4474F71F-26B0-411B-A3A2-16537132387F}" />
</Allergies>
<HealthIssueCodes>
<HealthIssueCode code="N17" codeType="ICD10" />
<HealthIssueCode code="M10" codeType="ICD10" />
<HealthIssueCode code="J45" codeType="ICD10" />
</HealthIssueCodes>
<DuplicateTherapy checkSameDrug="true" />
<DuplicateIngredient checkSameDrug="true" />
</References>
</Interaction>
<PatientProfile>
<Gender>F</Gender>
<Age><Year>34</Year></Age>
<Pregnancy><Week>20</Week></Pregnancy>
<Nursing>true</Nursing>
</PatientProfile>
</Request>
```

MIMS TYPE

MIMS GUID

Picking “MIMS TYPE” and “MIMS GUID” of all the drug being prescribed to Patient

Picking “MIMS TYPE” and “MIMS GUID” of all the drug allergies from the Patient

Patient’s ICD10 conditions

Default Duplicate Therapy and Duplicate Ingredient Tags

You need to add the Patient Profile tag

You need to add the patient profile tag as well as Nursing

Drug Interaction (20)

Allergy (3)

Health Conditions (3)

Duplicate Therapy (4)

Duplicate Ingredient (2)

Pregnancy (8)

Lactation (8)

- | |
|---|
| <p>amiodarone hydrochloride 150mg per 3mL Injection (amiodarone hydrochloride/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with extreme caution.]</p> |
| <p>Cordarone inj 150 mg/3 mL (amiodarone hydrochloride/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with extreme caution.]</p> |
| <p>amiodarone hydrochloride 150mg per 3mL Injection (amiodarone hydrochloride/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Avoid combination.]</p> |
| <p>Cordarone inj 150 mg/3 mL (amiodarone hydrochloride/Systemic) vs fluconazole 150mg Oral Tablet (fluconazole/Systemic) [Avoid combination.]</p> |
| <p>warfarin 2mg Oral Tablet (warfarin/Systemic) vs ethambutol hydrochloride - isoniazid - pyrazinamide - rifampicin Oral Tablet (rifampicin/Systemic) [Use combination with caution.]</p> |

Drug Interaction (20)

Allergy (3)

Health Conditions (3)

Duplicate Therapy (4)

Duplicate Ingredient (2)

Pregnancy (8)

Lactation (8)

- ▼ amiodarone hydrochloride 150mg per 3mL Injection (amiodarone hydrochloride/Systemic) vs warfarin 2mg Oral Tablet (warfarin/Systemic) [Use combination with extreme caution.]

Interaction Effect

amiodarone hydrochloride increases effect of warfarin

Severity Level

Severe - The interaction between these medications may be life-threatening or may cause permanent damage. These medications are not usually used concurrently; medical intervention may be required.

Documentation Level

Well Established - There have been several published reports of this interaction. The pharmacological explanation of why the interaction occurs is well documented and understood. There are usually controlled studies that have established that the interaction exists.

Probable Mechanism

Amiodarone may increase the anticoagulant effects of vitamin K antagonists by inhibiting their CYP450-mediated metabolism in a delayed manner but may persist for months after discontinuation of amiodarone. Bleeding incidents and increased bleeding risk and/or INR have been documented during combination of warfarin and amiodarone. The anticoagulant effects of phenprocoumon and acenocoumarol have also been potentiated by amiodarone. A reduction in oral anticoagulant dose of up to 60% may be required. Closely monitor patient's INR and observe for signs of bleeding, especially when amiodarone-induced thyrotoxicosis is present.

Actions to be taken

1. Use combination with extreme caution.
2. Decrease drug dosage.
3. Monitor for signs of drug toxicity.
4. Monitor INR.

References

- Caraco Y, Chajek-Shaul T. The incidence and clinical significance of amiodarone and acenocoumarol interaction. *Thromb Haemost.* 1989; 62(3/Nov 24):906-8
- Kurnik D, Loebstein R, Farfel Z et al. Complex drug-drug-disease interactions between amiodarone, warfarin, and the thyroid gland. *Medicine (Baltimore).* 2004; 83(2/Mar):107-13
- Cheung B, Lam FM, Kumana CR. Insidiously evolving, occult drug interaction involving warfarin and amiodarone. *BMJ.* 1996; 312(7023)(Jan 13):107-8
- Sanoski CA, Bauman JL. Clinical observations with the amiodarone/warfarin interaction: dosing relationships with long-term therapy. *Chest.* 2002; 121(1/Jun):19-23
- Vitry AJ, Roughhead EE, Ram say EN et al. Major bleeding risk associated with warfarin and co-medications in the elderly population. *Pharmacoepidemiol Drug Saf.* 2011; 20(10/Oct):1057-63
- Coumarins and related drugs + Amiodarone. Stockleys Drug Interactions. MedicinesComplete. Baxter K (ed). RPS Publishing. London. UK. Available from URL: <http://www.medicinescomplete.com/mc>. (Accessed on 19/02/2014)